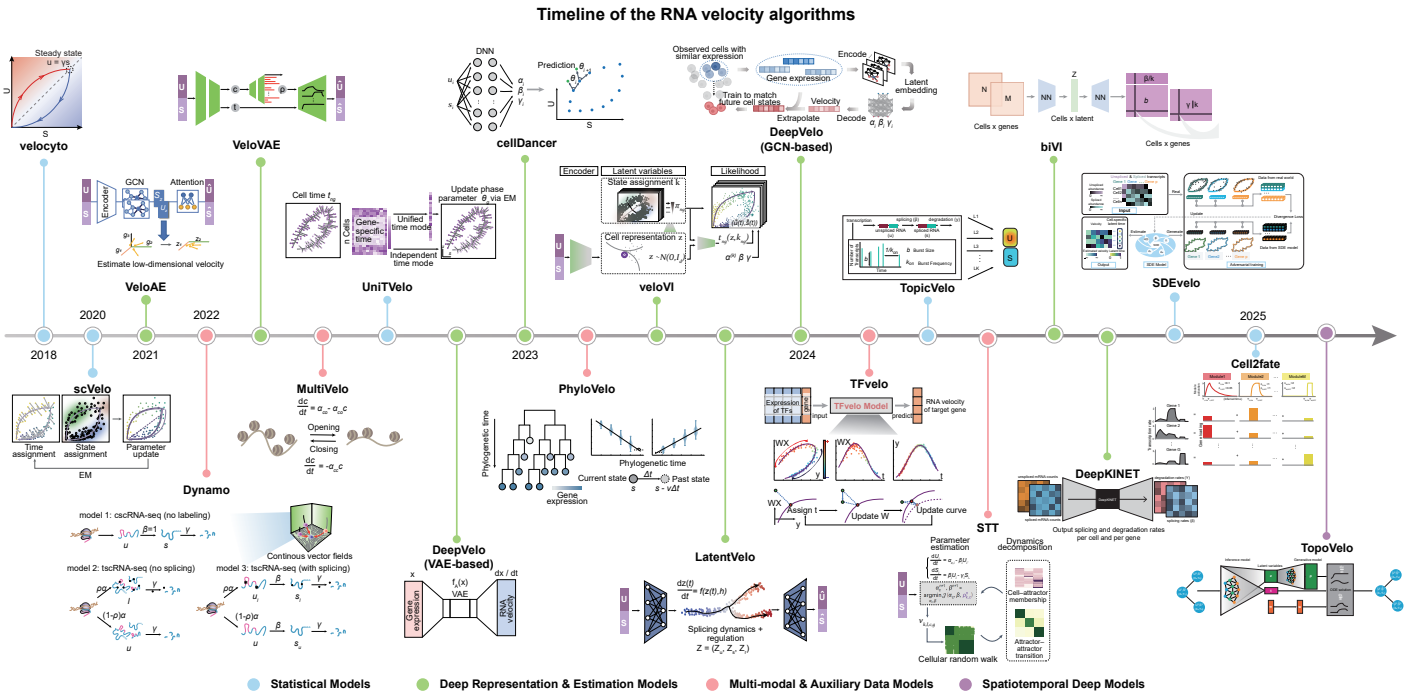


Fig. 1 | Chronological landscape of RNA velocity algorithms. Timeline of the 21 methods covered by the taxonomy, ordered by first online publication date from 2018 to 2025. Schematics for each method are positioned along the timeline to highlight key methodological and architectural innovations. Methods are classified into four categories according to their mathematical frameworks and input data requirements. See Methods and Supplementary Table 1 for detailed technical specifications.